CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of M. Sc. (Biotechnology) Backlog Examination, March 2019

Course code & Name: MS 811 & Omics and Bioinformatics

Date: 14/03/2019 Day: Thursday Time: 1:30 P.M. To 2:00 P.M. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	**Choose the correct answer for the following questions.**		**20**
1.	The following genome sequencing method requires generation of single stranded DNA.		01
	(i) Sanger sequencing	(ii) Oxford nanopore	
	(iii) 454 Pyrosequencing	(iv) All of them	
2.	The following technique for physical mapping of genomes utilizes irradiation to fragment the genomes but does not involve generation of somatic cell hybrids.		01
	(i) Hybridoma	(ii) Radiation Hybrid	
	(iii) HAPPY mapping	(iv) FISH mapping	
3.	The following technique is used to fabricate DNA microarrays.		01
	(i) Contact printing	(ii) Non-contact printing	
	(iii) Photolithography	(iv) All of them	
4.	In eukaryotic genome organization, the following sequences can be classified as dispersed repeats.		01
	(i) Simple sequence repeats (SSR)	(ii) Retroposons	
	(iii) Microsatellite	(iv) Minisatellite	

5.	Which of the following enzyme is used in pyrosequencing reaction mixtures.		01
	(i) ATP sulfurylase	(ii) Luciferase	
	(iii) DNA Polymerase	(iv) All of them	
6.	The following quantitative proteomics method utilizes ($^{2}H/^{13}C/^{15}N$) amino acids in cell culture medium for labeling proteins.		01
	(i) Protein microarrays	(ii) Isotope-coded affinity tag	
	(iii) Metabolic stable isotope coding	(iv) Enzymatic stable isotope coding	
7.	Large segments of the mouse and human genomes have closely related genes aligned in the same order on chromosomes, this relationship is known as		01
	(i) Synteny	(ii) LINES	
	(iii) Synchrony	(iv) SINES	
8.	In the yeast-two hybrid system, the cDNA sequences from a library are cloned in frame with coding sequences of either the DNA-binding domain or the activation domain of _______ transcription factor.		01
	(i) LacZ	(ii) LEU2	
	(iii) HIS3	(iv) Gal4	
9.	The following is a RefSeq database accession number format.		01
	(i) NM_123456	(ii) A1B2C3	
	(iii) ABC1234	(iv) All of them	
10.	In proteome analysis, the matrix assisted laser desorption ionization (MALDI) source coupled with a time of flight (TOF) analyzer is used for ___________.		01
	(i) Isoelectric focusing	(ii) Protein microarrays	
	(iii) Peptide mass fingerprinting	(iv) cDNA sequencing	
11.	The following groups of enzymes mediate functional group modifications of most of the drugs and the genes that encode them are highly polymorphic.		01
	(i) ATRA	(ii) MDR-1	
	(iii) Telomerase	(iv) Cytochrome P450	
12.	The following is an *in vivo* library-based screening methods allowing the large-scale analysis of binary protein-protein interactions.		01
	(i) Isotope coded affinity tag (ICAT)	(ii) Yeast two-hybrid system	
	(iii) Phage Display	(iv) Tandem affinity purification	
13.	The following evolutionary model assumes equal probability of substitutions (transitions and transversions) for all nucleotides to compute evolutionary distances.		01
	(i) Jukes–Cantor	(ii) Kimura	
	(iii) Bootstrapping	(iv) Maximum Likelihood	
14.	The following program is used to evaluate proteins structures.		01
	(i) VADAR	(ii) DSSP	
	(iii) PROCHECK	(iv) All of them	
15.	In the following technique proteins from two different samples are differentially labeled with fluorescent dyes and then separated on the same 2D polyacrylamide gel.		01
	(i) MudPIT	(ii) 2D-LC	
	(iii) 2D-DIGE	(iv) GC-MS	

16.	The following is a character-based method for constructing phylogenetic trees.		01
	(i) Neighbor Joining	(ii) Maximum Parsimony	
	(iii) Minimum Evolution	(iv) UPGMA	
17.	The following procedure is designed to capture multi-protein interaction clusters, followed by identification all the interacting proteins in the complex using mass spectrometry.		01
	(i) Phage display system	(ii) CE-MS	
	(iii) Yeast two hybrid system	(iv) Tandem affinity purification	
18.	In 2D-PAGE, during isoelectric focusing on an immobilized pH gradient (IPG) strip, a protein in a pH region below its isoelectric point (pI) will migrate towards?		01
	(i) Anode	(ii) Cathode	
	(iii) Stop migrating	(iv) All of them	
19.	The following scheme is used for classification of protein structures.		01
	(i) SCOP	(ii) CATH	
	(iii) PROCHECK	(iv) Both (i) and (ii)	
20.	The following comparative genomics method is used to infer protein-protein interactions directly from genomic data.		01
	(i) Rosetta stone	(ii) FRET	
	(iii) Epistasis	(iv) Gain of function mutations	

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III Semester of M. Sc. (Biotechnology) Backlog Examination, March 2019

Course code & Name: MS 811 & Omics and Bioinformatics

Date: 14/03/2019 Day: Thursday Time: 2:00 P.M. To 4:30 P.M. Maximum Marks: 50

Instructions:

1. **Section I and II must be attempted in TWO ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q – II Answer the following questions as directed (answer any five).		**20**
1.	Explain the pyrosequencing reaction chemistry and method.	04
2.	Explain the FISH-mapping of genomes.	04
3.	Explain the enrichment of metagenome for desired applications.	04
4.	Explain the shotgun genome sequencing approach.	04
5.	Give an overview of different types of repetitive DNA in eukaryotic genome organization.	04
6.	Explain the high throughput illumina-reversible terminator sequencing technology.	04
7.	Explain the photolithography method used for manufacturing DNA microarrays.	04
SECTION - II		
Q-III Answer the following questions as directed (answer any ten).		**30**
1.	Explain yeast-two hybrid system for screening protein-protein interactions	03
2.	Explain the 2D-PAGE and peptide mass-fingerprinting for identification of differentially expressed proteins.	03
3.	Explain the enzymatic stable isotope coding strategy for quantitative proteome analysis.	03
4.	Explain the ICAT method of quantitative proteomics.	03
5.	Describe different analytical techniques used in metabolomics.	03
6.	Write a short note on protein microarrays.	03
7.	Explain the phage display method to screen protein-protein interactions.	03
8.	Describe the nomenclature used to infer phylogenetic trees.	03
9.	Differentiate between protein motifs and domains.	03
10.	Differentiate between local and global sequence alignments.	03
11.	Describe the different types of biological databases giving examples.	03
12.	Expand the following terms: (i) NCBI (ii) PHYLIP (iii) PDB	03